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A novel TCF calibration method for robotic visual measurement system

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ABSTRACT

An accurate TCF (tool control frame) model is essential for high-accuracy robot off-line programming. Meanwhile, TCF calibration is an important procedure for production recovery after robot collides in industrial field. This article proposes a novel TCF calibration method in robotic visual measurement system in which the robot TCF is defined based on the model of visual sensor and a standard sphere with known diameter is utilized as the calibration target. With the translational and rotational movements of the industrial robot, the visual senor measures the center of standard sphere from multiple different robot postures, TCF orientation and TCP position are determined in two steps. Robot off-line programming is performed based on the TCF calibration result, and robot collision is simulated on an ABB IRB2400 industrial robot. Experimental results have validated the effectiveness and efficiency of the standard sphere-based TCF calibration method, which could control the deviation of an identical featured point within 0.5 mm measured before and after collision recovery.

1. Introduction

Product quality inspection and controlling system is of great significance to improve the comprehensive competence of the modern enterprises. Robotic visual measurement system that takes advantage of the industrial robot (high flexibility, high repeatability, large motion range) and visual sensor (high-accuracy, non-contact measurement) has been extensively applied for quality inspection in the production assembly lines. The automobile body-in-white inspection system based on industrial robot is a case in point [1,2], in which the industrial robot is utilized as the orienting device and drives the visual sensor to inspect the key dimensions on the car body-in-white. For the identical products with a same model on the assembly line, the industrial robot follows a same path for inspection. Owning to the high repeatability of the industrial robot, the key geometry dimensions of products on the assembly line could be 100% inspected and deviations could be detected. Generally, industrial robot off-line programming is performed before the robotic repetitive movement. And in the off-line programming of the robotic visual measurement system, an optimal robot trajectory is planned according to the distribution of the featured points that need to be inspected [3]. Additionally, there is an optimal measurement range in the visual sensor usually, within which a best measuring accuracy could be obtained [4]. As the visual sensor serves as the end-effector of the industrial robot, it is important for the robot to drive it to the optimal working position. Hence, an accurate model of the end-effector is critical for the high-accuracy robot off-line programming, without which even the most accurate robot will fail to complete high-accuracy robotic measurement. Meanwhile, for the robot on-line teaching programming, an end-effector model that matches with human's operating habits will make the path planning more convenient and more labor-saving. TCF (tool control frame) is used to define the end-effector in the robot end-flange frame. As the accuracy of the end-effector model directly affects the robot motion accuracy, an accurate and feasible TCF calibration method is of great significance in robotic applications.

Additionally, as the robotic visual measurement system works in the long time, affected by the environment and applied load, occurrence of robot collision is unavoidable. This will directly result in suspension of the production line, delay of production cycle and economic losses. In this case, replacement of the visual sensor or mechanical adapter immediately is required for the resumption of production line. As there is difference among the intrinsic and extrinsic parameters of visual sensors, also in the structure of mechanical adapters, the position and orientation of TCF with respect to the robot end-flange will be changed after visual sensor or adapter replacement. Even a small change will cause inaccuracy in the end-effector model, as well as inaccuracy of the robot

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trajectories. The small change of the end-effector model may cause the featured points beyond the optimal measurement range of the visual sensor, which will result in inaccurate measurement, or even cause the laser sensor unable to measure the featured points. In order to avoid the laborious task of re-teaching the robot trajectory, re-calibration of the TCF is requisite to recover the robotic inspection efficiently after robot collision.

The vast majority of the existing studies focus on the TCF calibration of the robot end-effector such as welding torch, drilling tools and so on. Some researches propose many restrictions about the geometry of the tools or the calibration process. TCF calibration method proposed by X. Wang and Edward Red requires that the tools should be an axially symmetric rotating body and the axis should coincides or parallel with the axis of the robot's last joint when installed [5,6]. Some other scholars have successfully calibrated the TCF by adopting external device. Wu L. and Yang X., etc. calibrate the robot TCF based on a planar template. Cheng F S recovers position of TCF origin by employing a special external measuring system (DynaCal from Dynalog Inc.) [7,8]. But these TCF calibration methods are expensive and time-consuming. In the case of visual sensor serving as the robot end-effector, many researchers define the TCF the same as the sensor frame and simplify the TCF calibration to hand-eye calibration. The hand-eye calibration is to solve the homogeneous equation AX = XB or AX = YB where X and Y are the unknown hand-eye transformation and the robot-world transformation, A and B are the movement of robot end-flange and camera [9,10]. Several hand-eye calibration methods including linear solution and nonlinear solution were reported in literatures [11,12]. But the TCF definition in this method does not match with the human operation habits. Moreover, it involves complex computation and singular solution may be obtained.

For the robotic visual measurement system, TCF is defined based on the model of visual sensor in this paper. Then a novel sphere-based TCF calibration method is proposed in detail. In order to determine the orientation of the TCF, the robot moves in translational motion mode, and the line structured laser sensor measures a standard sphere at multiple robot position. Then by controlling the robot posture and making the line structured laser sensor observe the standard sphere from several different directions, the TCP (tool center point) position could be determined. In this paper, a robotic visual measurement system is built with an ABB IRB2400 robot and a line structured laser sensor, TCF calibration is performed with the method proposed in this paper. And then robot off-line programming is performed to verify the effectiveness and accuracy of this method. Through simulation of robot collision, robotic inspection recovery is validated by re-calibrating the TCF. Experiment results have shown that the method is convenient, reliable and could serve as an effective and high-accurate solution for TCF calibration in the industrial field.

The structure of this paper is arranged as follows: Section 2 introduces the principle of the robotic visual measurement system and then reviews the traditional TCF calibration method in robotic visual inspection system; Section 3 presents the algorithm schematic for the sphere-based TCF calibration method; experiments are designed in Section 4 for verification, analysis of the effectiveness of the TCF calibration method; Section 5 concludes the article.

2. Background knowledge

2.1. Principle of robotic visual measurement system

The robotic visual measurement system mainly consists of an industrial robots and a line structured laser sensor. As shown in Fig. 1, the coordinate systems of the robotic visual inspection system consist of robot base frame {BF}, robot end-flange frame {EF}, laser sensor frame {SF} and workpiece frame {WF}. Measurement model of the system is expressed as follows:

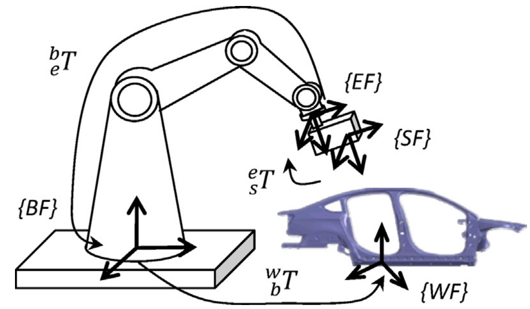


Fig. 1. Schematic components of robotic visual measurement system.

$$P_w = {}^{w}_{b}T \times {}^{b}_{e}T \times {}^{e}_{s}T \times P_s \tag{1}$$

where P_s and P_w are the coordinates of a featured point P in laser sensor frame and workpiece frame respectively. ${}^{e}_{s}T$, ${}^{b}_{e}T$ and ${}^{w}_{b}T$ are 4×4 homogeneous coordinate transform matrices. ${}^{e}_{s}T$ is the coordinate transform relationship between the laser sensor frame and the end-flange frame, which is also known as hand–eye relationship; ${}^{b}_{e}T$ is coordinate transform relationship between robot end-flange frame and the base frame which could be obtained from robot forward kinematics; ${}^{w}_{b}T$ is coordinate transform relationship between robot base frame and workpiece frame. The measured results of the workpiece are transformed and assessed in the workpiece frame as well.

The line structured laser sensor is characterized by its high-accuracy, compacted-sized, real-time processing and easy-operation, and it has been widely applied in industrial on-line inspection. The line structured laser sensor works on the optical triangulation principle, as shown in Fig. 2 [13,14]. The measured object is illuminated from one direction by a laser projector and viewed by the camera from another. The exact height of each spot on the laser line can be calculated by intersecting the corresponding camera projection ray with sheet of laser. The camera frame O_c-$x_c y_c z_c$ is defined as the global frame of the laser sensor in this paper, where O_c is the camera optical center, x_c, y_c axes are in the same direction with pixel column and row increasing direction in the camera image plane, z_c axis is the camera optical axis. Camera image plane π_n is perpendicular to the optical axis with a distance of f away from the optical center O_c. The image plane frame is O_n-$x_n y_n$ and the laser sheet plane that projected by the laser emitter is plane π_s. Assuming $P_s(x, y, z)$ is a visual point on the laser plane π_s, it can be expressed in laser sensor frame as follows:

$$ax + by + cz + d = 0 \tag{2}$$

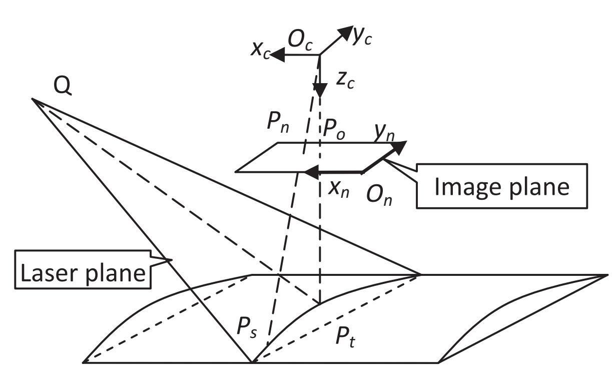


Fig. 2. Measurement model of line structured laser sensor.

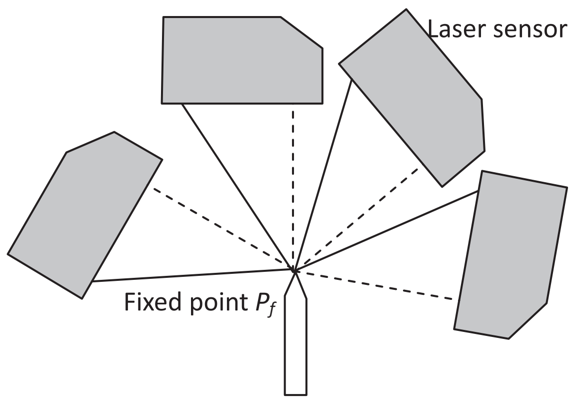


Fig. 3. TCP calibration based on fixed reference point.

where (a, b, c, d) are structural parameters of the laser plane. Based on the camera perspective projection principle, P_s is projected at P_n on the image plane, then O_c, P_s and P_n will be collinear and the linear equation can be given as:

$$\frac{x}{x-u} = \frac{y}{y-v} = \frac{z}{z-f} \tag{3}$$

where (u, v, f) are coordinate of P_n in the sensor frame. Simultaneously solving Eqs. (2) and (3), we can obtain coordinates of the point P_s in the sensor frame.

2.2. Conventional TCF calibration method

In the robotic visual measurement system, the line structured laser sensor is mounted on the robot end-flange and serves as the end-effector. The robot trajectory is programmed based on the model of end-effector and the industrial robot drives the sensor to the right above of the featured points to be inspected. Robot TCF (tool control frame) is used to define the model of end-effector, which is made up of TCF orientation and TCP position. TCP locates at the TCF origin. TCF calibration is to determine the TCP position and TCF orientation relative to the robot end-flange frame, which are unknown but determinate [5]. The traditional way to calibrate the robot TCF in robotic visual measurement system learns from the robot TCP calibration method for tools such as welding torch, drilling tools and so on.

In the traditional ways, the orientation of TCF is defined the same as the robot end-flange frame, and only the TCP position is calibrated. By controlling the robot posture and making the line structured laser sensor observe a reference point fixed in robot volume from different directions, TCP position can be determined. As shown in Fig. 3, the robot is controlled to align the TCP to a reference point P_f fixed in the robot volume at several different robot postures, and position of TCP is solved with a least-square method.

Assuming that X_b is the coordinate of the reference point P_f in the robot base frame, R_i and T_i are orientation and position of robot end-flange, which can be obtained from robot forward kinematics. Line structured laser sensor is fixed at the robot end-flange as end-effector, assuming X_t is the TCP position in robot end-flange frame. When the robot is controlled to align the TCP to the reference point at several different robot postures, we can obtain the following equation:

$$X_b = R_1 \cdot X_t + T_1 = R_2 \cdot X_t + T_2 = \cdots = R_n \cdot X_t + T_n \tag{4}$$

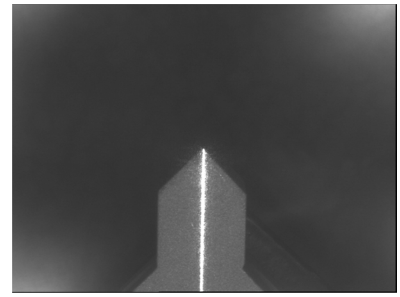


Fig. 4. Alignment of TCP with reference point.

Subtracting each two adjacent equations, an equation in form of $AX = B$ can be obtained as follows:

$$\begin{bmatrix} R_2 - R_1 \\ R_3 - R_1 \\ \vdots \\ R_n - R_1 \end{bmatrix} \cdot X_t = \begin{bmatrix} T_1 - T_2 \\ T_1 - T_3 \\ \vdots \\ T_1 - T_n \end{bmatrix} \tag{5}$$

As long as the coefficient matrix in Eq. (5) is nonsingular, X_t can be solved by means of least-square:

$$X_t = (A^T A)^{-1} A^T B \tag{6}$$

where $A = \begin{bmatrix} R_2 - R_1 \\ \vdots \\ R_n - R_1 \end{bmatrix}$, $B = \begin{bmatrix} T_1 - T_2 \\ \vdots \\ T_1 - T_n \end{bmatrix}$. The calibration error of TCP position can be calculated, which is the least-square fitting error:

$$\delta = \left\| \begin{bmatrix} R_2 - R_1 \\ \dots \\ R_n - R_{n-1} \end{bmatrix} \cdot X_t - \begin{bmatrix} T_1 - T_2 \\ \dots \\ T_{n-1} - T_n \end{bmatrix} \right\|$$
$$= \left(\sum_{i=1}^{n-1} |(R_{i+1} - R_i) \cdot X_t - T_i + T_{i+1}|^2 \right)^{1/2} \tag{7}$$

The purpose of the TCP alignment is to make the position of the robot TCP coincide with the reference point fixed in the robot volume. Based on the principle of perspective projection, all points on the optical axis are imaged at the principle point P_o on the image plane. If the image of the laser stripe also passes through point P_o, then P_o is the image point of TCP in the image plane. When the reference point P_f is also projected at the P_o, TCP is aligned with the external reference point in the physical world. So the judging criteria of the alignment of robot TCP with the reference point is that the reference point is imaged at the principle point on the image plane and the laser stripe passes through the principle point also. Spire is usually taken as the reference point in practical operations as shown in Fig. 4. However, there is never a spire that is sophisticated enough to be taken as a point, so it is difficult to guarantee the accuracy of the TCP position. Moreover, alignment of these three points is extremely time-consuming, with each alignment taking more than 5 min.

3. Sphere-based TCF calibration method

The traditional method presented in Section 2.2 defines the TCF orientation the same as robot end-flange frame, which does not match with the properties of structured laser sensor. In visual measurement system, it is expected that the camera in the laser sensor views the featured point head-on. In this paper, a novel sphere-based TCF calibration method is proposed, in which the TCF

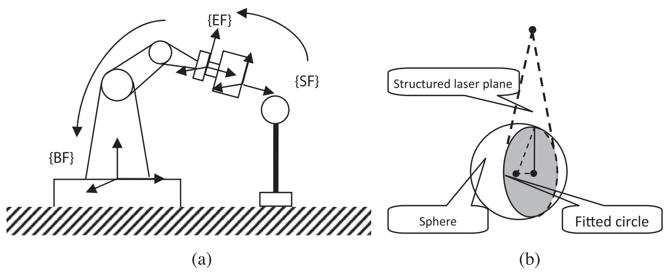


Fig. 5. Calibration of TCF based on standard sphere.

orientation is defined the same as the orientation of sensor frame O_c-$x_c y_c z_c$. As shown in Fig. 2, the camera optical axis direction is defined as the TCF z-axis direction, the x-axis and y-axis direction of sensor frame are defined as the TCF x-axis and y-axis direction. Meanwhile, as the optimal working position of the line structured laser sensor locates around the intersection point of camera optical axis z_s and the laser plane π_s, the point P_t in Fig. 2 is defined as the TCP position.

In order to calibrate the TCF orientation and the TCP position, a standard sphere with known diameter is fixed in the robot volume and applied as the calibration target as shown in Fig. 5a. The line structured laser sensor is driven to measure the sphere center at different robot postures. The laser sensor determines the position of the sphere center at each robot pose just by one laser shoot and the calculating method is shown in Fig. 5b. When the laser is projected on the sphere surface, an image of the arc shaped laser stripe is captured by the sensor and point cloud of the arc could be obtained by image processing. By means of the least-square fit, the circle center $Z(x_c, y_c, z_c)$ and circle radius r can be determined. The normal vector of the laser plane (a, b, c), circle center coordinate $Z(x_c, y_c, z_c)$ and sphere center coordinate $O(x_s, y_s, z_s)$ in laser sensor frame will satisfy the following relationship:

$$\begin{cases} x_s = x_c \pm at \\ y_s = y_c \pm bt \\ z_s = z_c \pm ct \end{cases} \tag{8}$$

where $t = \sqrt{R^2 - r^2}/\sqrt{a^2 + b^2 + c^2}$, R is the sphere radius, plus or minus is determined by the relative position between the circle center and the sphere center.

3.1. Determining TCF orientation

According to the TCF definition, TCF orientation is the same as the orientation of the laser sensor frame. So the orientation of the laser sensor frame is determined here.

The standard sphere measured by the line structured laser sensor is transformed to robot base frame according to the measurement model in Eq. (1). For a point P fixed in robot base frame, its coordinates $X_b = [x_b, y_b, z_b]^T$ in robot base frame and $X_s = [x_s, y_s, z_s]^T$ in laser sensor frame satisfy the following relationship:

$$\begin{bmatrix} X_b \\ 1 \end{bmatrix} = \begin{bmatrix} R_0 & T_0 \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} R_s & T_s \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} X_s \\ 1 \end{bmatrix} \tag{9}$$

where R_0, T_0 are the orientation and position of robot end-flange frame relative to robot base frame; R_s, T_s are the rotation matrix and translation matrix between laser sensor frame and end-flange frame which are to be determined. In an alternative way, Eq. (9) can be expanded:

$$X_b = R_0 \cdot R_s \cdot X_s + R_0 \cdot T_s + T_0 \tag{10}$$

Since the standard sphere is fixed in robot base frame, the coordinated of the sphere center is invariable in the robot base frame. If controlling the robot to drive the line structured laser sensor to measure the fixed point from two different robot poses, the following two equations can be obtained:

$$\begin{cases} X_b = R_{01} \cdot R_s \cdot X_{s1} + R_{01} \cdot T_s + T_{02} \\ X_b = R_{02} \cdot R_s \cdot X_{s2} + R_{02} \cdot T_s + T_{02} \end{cases} \tag{11}$$

If the orientation of robot end-flange remains unchanged during movement (robot translational movement), $R_{01} = R_{02} = R_0$. Subtracting the two equations in Eq. (11):

$$R_0 \cdot R_s \cdot (X_{s1} - X_{s2}) + T_{01} - T_{02} = 0 \tag{12}$$

Since R_0 is an orthogonal matrix, repeating the above procedure and we have:

$$\begin{aligned} & R_s \cdot [\, X_{s1} - X_{s2} \quad X_{s1} - X_{s3} \quad \cdots \quad X_{s1} - X_{sn} \,] \\ & \quad = R_0^T \cdot [\, T_{02} - T_{01} \quad T_{03} - T_{01} \quad \cdots \quad T_{0n} - T_{01} \,] \end{aligned} \tag{13}$$

If defining $A = [X_{s1} - X_{s2} X_{s1} - X_{s3} \cdots X_{s1} - X_{sn}]$, $B = R_0^T \cdot [T_{02} - T_{01} T_{03} - T_{01} \cdots T_{0n} - T_{01}]$. Eq. (13) can be simplified to a matrix equation in form of $R_s A = B$, and R_s can be solved by singular value decomposition (SVD):

$$R_S = VU^T \tag{14}$$

where V, U are right and left singular matrix of AB^T.

3.2. Determining TCP position

Against at weak robust and low precision of the conditional TCP calibration method proposed in Section 2.2, a nonlinear and high precision TCP calibration method based on spherical constraint is put forward in this section. With the fast development of modern machine tools, the accuracy of the standard sphere has attained the level of micron or even submicron.

If controlling the robot to align the TCP onto the surface of the standard sphere at different robot poses, we have:

$$X_{b1} = R_1 \cdot X_t + T_1, \quad X_{b2} = R_2 \cdot X_t + T_2, \cdots, \quad X_{bn} = R_n \cdot X_t + T_n \tag{15}$$

where X_{bn} is the coordinated of the TCP in the robot base frame. As the TCP is aligned on the surface of the sphere in different robot poses, X_{b1}, X_{b2}, X_{bn} are the coordinates of the sample points on the standard sphere in the robot base frame.

Suppose the coordinate of sphere center in the robot base frame is $X_p = (x_p, y_p, z_p)$ and the radius of the standard sphere is R, when the TCP is aligned on the surface of the sphere, we have:

$$|X_{bi} - X_p| = R \tag{16}$$

As X_{b1}, $X_{b2}, \ldots$, X_{bn} are sample points on the surface of the standard sphere, the following objective function can be obtained:

$$f = \sum_{i=1}^{n} ((X_{bi} - X_p) \cdot (X_{bi} - X_p)^T - R^2) \tag{17}$$

As the radius of the standard is known with a high degree of precision, the parameters to be optimized are X_t and X_p. All the parameters could be optimized iteratively by using a Levenberg–Marquardt nonlinear optimization.

As is discussed in the foregoing section, when the image of the laser stripe passes through the principle point P_o, P_o is the image point of TCP on the image plane. So the judging criteria of the alignment of robot TCP with the surface of the standard sphere is that the

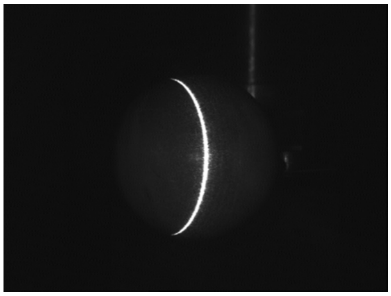


Fig. 6. Alignment of TCP with the sphere surface.

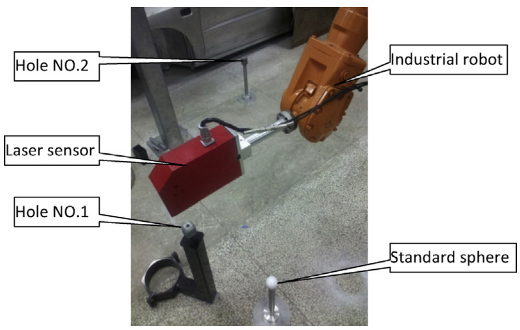


Fig. 7. Experimental setup for robotic visual measurement system.

laser stripe passes through the principle point on the image plane and the laser stripe is aligned on the surface of the standard sphere in the physical world at the same time, as shown in Fig. 6. This is much easier to achieve than alignment of the TCP to a reference point as shown in Fig. 4. It takes less than 1 min for each alignment, so the operation staff will benefit greatly from this method. The pixel coordinate of the principle point is obtained in camera calibration.

For a general industrial robot, geometric parameter deviation, elastic deformation, gear clearance and other factors create significant nonlinear position errors, that is, the robot end-effector position errors are different in different robot positions and postures. In order to minimize the impact of position error, the robot movements are restricted in a small volume by just rotating the joints 4, 5, and 6.

4. Experiment and analysis

In this article, a robotic visual measurement system is constructed by mounting a line structured laser sensor on an ABB IRB2400/10 industrial robot as shown in Fig. 7. Repeatability of the IRB2400/10 industrial robot is 0.06 mm, measurement accuracy and repeatability of the line structured laser sensor are ±0.1 mm and 0.02 mm respectively. Ten featured holes are distributed uniformly in the robot volume, and the hold centers are the features to be inspected. TCF recovery experiment is performed through simulated robot collision. In order to simulate the tool deformation caused by robot collision in the industrial field, the mechanical adapter is replaced by another one which is slightly different in size. The feasibility and versatility of the TCF calibration method are analyzed based on the results of the featured holes.

As ABB robots represent the orientation in quaternion, which can be converted to a rotation matrix by the following equation:

$$R = \begin{bmatrix} {q_0}^2+{q_1}^2-{q_2}^2-{q_3}^2 & 2(q_1q_2-q_0q_3) & 2(q_1q_3+q_0q_2) \\ 2(q_1q_2+q_0q_3) & {q_0}^2+{q_2}^2-{q_1}^2-{q_3}^2 & 2(q_2q_3-q_0q_2) \\ 2(q_1q_3-q_0q_2) & 2(q_2q_3+q_0q_2) & {q_0}^2+{q_3}^2-{q_1}^2-{q_2}^2 \end{bmatrix} \qquad (18)$$

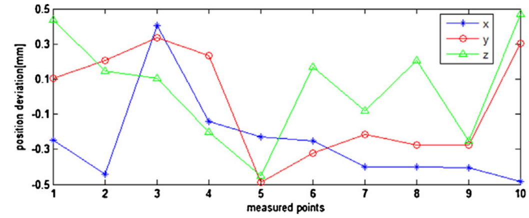


Fig. 8. Deviation of test points after tool recovery based on standard sphere.

where q_0, q_1, q_2, q_3 are the four elements for the robot orientation. Accordingly, the TCF orientation should be converted to quaternion in the robot program.

A standard sphere with a radius of 19.1 mm is fixed in robot volume. Controlling the robot in translational motion mode and making the line structured laser sensor measure the sphere center at four different positions, TCF orientation Rs can be solved as presented in Section 3.1. Then controlling the robot in rotational motion mode and locating the TCP on the sphere surface form six different robot postures, TCP position can be solved according to Eq. (17). Then *tool*1 parameters are calculated as follows:

$$R_{s1} = \begin{bmatrix} -0.87736 & 0.00291 & 0.47983 \\ 0.00761 & 0.99994 & 0.00785 \\ -0.47978 & 0.01054 & -0.87733 \end{bmatrix},$$

$$T_{s1} = [\,275.724 \quad -1.621 \quad 357.464\,]^T,$$

$$tool1 = [\,275.724 \quad -1.621 \quad 357.464\,], \\ [\,0.2476 \quad 0.0027 \quad 0.9688 \quad 0.0047\,].$$

Robot teaching programming is performed based on *tool*1 in which the visual sensor is moved to keep the center of featured holes as close as the TCP positions in different robot postures. Centers of the ten featured holes are measured and recorded. Replacing the mechanical adapter to simulate the robot collision and the TCF is re-calibrated with the sphere-based TCF calibration method. Changing the tool parameters in the robot program according to TCF calibration result, the centers of ten featured holes are re-measured without re-teaching the robot path. TCF parameters and *tool*2 parameters calibrated after simulated robot collision are:

$$R_{s2} = \begin{bmatrix} -0.88601 & 0.00134 & 0.46366 \\ 0.00526 & 0.99996 & 0.00716 \\ -0.46363 & 0.00878 & -0.88599 \end{bmatrix},$$

$$T_{s2} = \left[\,275.796 \quad -1.532 \quad 346.292\,\right]^T,$$

$$tool2 = [\,275.724 \quad -1.621 \quad 357.464\,], \\ [\,0.2476 \quad 0.0027 \quad 0.9688 \quad 0.047\,].$$

Re-defining the TCF parameters in the robot program and the robot trajectory will adjust by itself. Control the robot run the adjusted robot program and drive the sensor to measure the ten featured hole. If ignoring the measurement error of the laser sensor, the deviation of measured results of featured holes in laser sensor frame reflects the deviation of position and orientation of

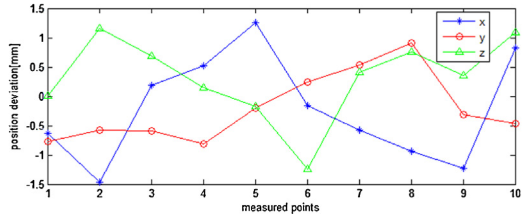
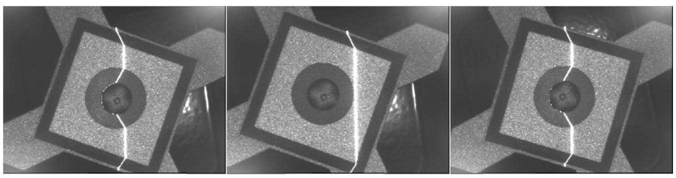


Fig. 9. Deviation of test points after TCP recovery based on reference point.



(a) before collision (b) after collision (c) after recovery

Fig. 10. Images of a test point in robot collision.

laser sensor frame, which indicates the accuracy of the TCF calibration also. The deviation between the measured hold centers before and after simulated robot collision is show in Fig. 8.

For comparison, the experiment is repeated based on the traditional TCF calibration method presented in Section 2.2. And the deviation between the measured hold centers before and after simulated robot collision is show in Fig. 9.

It could be seen from Figs. 8 and 9 that both TCF calibration methods are capable of calibrating the TCF parameters and realizing measurement recovery after robot collision. Measured results with the sphere-based TCF calibration method depict a much smaller deviation after a simulated collision which means model of TCF is more accurate calibrated with the sphere-based TCF method. Fig. 10 shows the images of the hole-center NO. 4 before simulated collision, after simulated collision, and after TCF recovery with sphere-based TCF calibration method respectively. It can be seen, when robot collides, the laser stripe projected by the line structured laser sensor is away from measured features obviously and he laser sensor is not able to measure the hole-center if the robot is not re-programmed. With the sphere-based TCF re-calibration, the laser stripe almost moves back to the same position as before and it can be seen from Fig. 8 that coordinate of hole-center NO. 4 deviates is less than 0.3 mm. The robot trajectory will adjust by itself after re-defining the TCF parameters in the robot program. So when robot collision occurs, only the TCF calibration is performed to determine the TCF orientation and TCP position after sensor or adapter changed, and there is no need to re-teach the robot program.

The conventional TCF calibration method depends on the human eye to determine whether the imaged TCP position is aligned to the reference point which is time-consuming and limited to a certain precision. It takes about 20 min to align the TCP to the spire from four different robot poses in the experiment. The TCF alignment in sphere-based method is much more convenient and the whole TCF calibration process takes less than 10 min. On the other hand, sphere-based TCF calibration method can also extent its applications to TCP calibration for tools such as grippers, welding torch and probe sensors.

5. Conclusions

Accurate robot off-line programming can only achieve with accurate TCF model, and TCF calibration is also an important procedure in robot collision recovery. This paper proposes a simple but feasible sphere-based TCF calibration method for robotic visual measurement system in which the visual sensor severs as the end-effector. Calibration experiment and robot off-line programming are performed on an ABB IRB2400/10 industrial robot. Experimental results show that the proposed method can serve as high-efficiency, reliable solutions for TCF calibration which could complete the calibration process in less than 10 min. Experiment of simulated robot collision also shows that TCF calibration method based on standard sphere can control the deviation measured before and after the collision less than 0.5 mm. Since it is easy and convenient to implement, we expected the proposed TCF calibration methods will have its wide application in the industrial field for robot on-line accuracy enhancement and robot collision recovery.

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